

WASP-12b

R-0553

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_180116 · created 2026-07-22T11:04:00+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458134.67236 +/- 0.00046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1245 +/- 0.0064
Transit depth (Rp/Rs)^2	1.55 +/- 0.16 [%]
Orbital Inclination (inc)	79.54 +/- 1.52
Airmass coefficient 1 (a1)	1.016 +/- 0.0082
Airmass coefficient 2 (a2)	-0.0127 +/- 0.0035
Scatter in the residuals of the lightcurve fit is	0.81 %
Best Comparison Star	#3 - [356, 239]
Optimal Aperture	4.15
Optimal Annulus	6.47
Transit Duration (day)	0.1154 +/- 0.0051

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1245 ± 0.0064 vs 0.1178 (deviation 0.85σ)
Duration fit vs published	0.1154 d vs 0.125 d (deviation 1.53σ)
Mid-time O–C	-1.4 min
Depth SNR	15.8
Residual scatter	0.81 %

Light curve

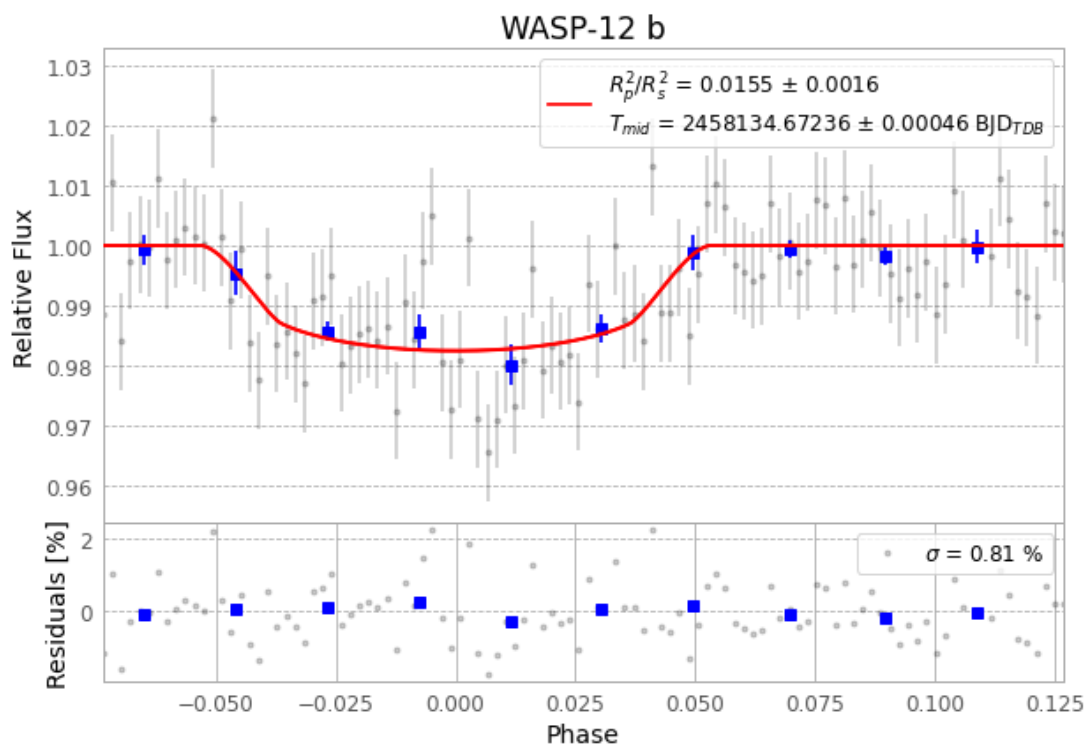


Figure 1 — FinalLightCurve_WASP-12 b_2018-01-15.png (SHA-256 28a31b6a156b58aa...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [373, 208], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a26ab3d16b7fbd28...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

108 FITS frames (71 MB), WASP-12180116015712.FITS → WASP-12180116071812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-180116.log (0688c06335db...), FinalLightCurve_WASP-12 b_2018-01-15.png (28a31b6a156b...), FinalLightCurve_WASP-12 b_2018-01-15.csv (c8613f65b2bb...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-22 11:04Z.